MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02 (SOLUTIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The first four terms of an AP whose first term is -2 and common difference is -2 are: [1]

(A) $-2, 0, 2, 4$ (B) $-2, 4, -8, 16$ (C) $-2, -4, -6, -8$ (D) $-2, -4, -8, -16$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) $-2, -4, -6, -8$** [0.5 Mark]

Given: First term $a = -2$, common difference $d = -2$. We know the standard form of an AP:

Formula / Theorem Used: $a,\ a+d,\ a+2d,\ a+3d,\ \ldots$

Justification: [0.5 Mark]

$a_1 = -2$; $a_2 = -2 + (-2) = -4$; $a_3 = -4 + (-2) = -6$; $a_4 = -6 + (-2) = -8$.

Q2. The 21st term of the AP whose first two terms are -3 and 4 is: [1]

(A) 17 (B) 137 (C) 143 (D) -143

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 137** [0.5 Mark]

Given: $a_1 = a = -3$, $a_2 = 4 \implies d = a_2 - a_1 = 4 - (-3) = 4 + 3 = 7$.

We know the general term formula of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$$a_{21} = -3 + (21 - 1) \times 7 = -3 + 20 \times 7 = -3 + 140 = \mathbf{137}$$

Q3. If the 2nd term of an AP is 13 and the 5th term is 25, what is its 7th term? [1]

(A) 30 (B) 33 (C) 37 (D) 38

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(B) 33** [0.5 Mark]

We apply $a_n = a + (n-1)d$:

Formula / Theorem Used: $a_2 = a + d = 13$ and $a_5 = a + 4d = 25$

Justification: [0.5 Mark]

Subtracting $a + d = 13$ from $a + 4d = 25$: $3d = 12 \implies d = 4$, and $a = 13 - 4 = 9$.

Therefore: $a_7 = a + 6d = 9 + 6(4) = 9 + 24 = \mathbf{33}$.

Q4. Which term of the AP $21, 42, 63, 84, \ldots$ is 210? [1]

(A) 9th (B) 10th (C) 11th (D) 12th

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(B) 10th** [0.5 Mark]

Given: $a = 21$, $d = 42 - 21 = 21$, $a_n = 210$. Using the nth term formula:

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$$21 + (n-1)(21) = 210 \implies 21n = 210 \implies n = \frac{210}{21} = \mathbf{10}$$

Q5. **Assertion (A):** The nth term of the AP $a, a+d, a+2d, \ldots$ is given by $a_n = a + (n-1)d$.
Reason (R): Each term is obtained by adding the common difference d to the preceding term. [1]

(A) Both (A) and (R) are true; (R) explains (A). (B) Both are true; (R) does not explain (A).
(B) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true; (R) explains (A).** [0.5 Mark]

We examine how the general term arises from the recursive progression definition:

Formula / Theorem Used: $a_1 = a,\ a_2 = a + d,\ a_3 = a + 2d,\ \ldots,\ a_n = a + (n-1)d$

Justification: [0.5 Mark]

To reach the nth term starting from $a_1 = a$, the common difference d is added $(n-1)$ times, directly producing $a_n = a + (n-1)d$. Hence both statements are true and (R) explains (A).

Q6. Which term of the AP $3, 15, 27, 39, \ldots$ will be 120 more than its 21st term? **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: AP is $3, 15, 27, 39, \ldots$, so $a = 3$ and $d = 15 - 3 = 12$.
We know the general term formula of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Finding the 21st Term [1 Mark]

$$a_{21} = 3 + (21-1) \times 12 = 3 + 20 \times 12 = 3 + 240 = 243$$

Step 2: Finding n Such That $a_n = a_{21} + 120$ [1 Mark]

$$a_n = 243 + 120 = 363 \implies 3 + (n-1) \times 12 = 363$$

$$12(n-1) = 363 - 3 = 360 \implies n - 1 = \frac{360}{12} = 30 \implies \boxed{\mathbf{n = 31^{st}\ \text{term}}}$$

Q7. Find the sum of all two-digit natural numbers which leave a remainder 1 when divided by 4. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Forming the Finite AP [1 Mark]

Numbers of the form $4k + 1$ in the two-digit range (10 to 99): Smallest $= 4(3) + 1 = 13$ (since $4(2) + 1 = 9$ is one-digit), largest $= 4(24) + 1 = 97$.
Thus the AP is $13, 17, 21, \ldots, 97$ with $a = 13$, $d = 4$, and $l = a_n = 97$.
We know the general term and sum formulas of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + l)$

Step 2: Finding the Number of Terms n [1 Mark]

$$97 = 13 + (n-1) \times 4 \implies 97 - 13 = 4(n-1) \implies 84 = 4(n-1)$$

$$n - 1 = \frac{84}{4} = 21 \implies n = 21 + 1 = 22$$

Step 3: Calculating the Sum S_{22} [1 Mark]

$$S_{22} = \frac{22}{2}(13 + 97) = 11 \times 110 = \boxed{\mathbf{1210}}$$

Q8. The angles of a quadrilateral are in AP. If the greatest angle is double the least angle, find all four angles of the quadrilateral. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let the four angles in ascending order be $a,\ a+d,\ a+2d,\ a+3d$ $(d>0)$.
We know the Angle Sum Property of a quadrilateral:

Formula / Theorem Used: Sum of interior angles of a quadrilateral $= 360°$

Step 1: Using Angle Sum and Greatest/Least Condition [1.5 Marks]
Sum $= 360° \implies a+(a+d)+(a+2d)+(a+3d) = 360° \implies 4a+6d = 360° \implies 2a+3d = 180°$ — (i)
Given greatest angle $= 2\times$ least angle:

$$a+3d = 2a \implies a = 3d \qquad \text{— (ii)}$$

Step 2: Solving for d, a, and All Four Angles [1.5 Marks]
Substituting $a = 3d$ into (i):

$$2(3d)+3d = 180° \implies 6d+3d = 180° \implies 9d = 180° \implies d = \frac{180°}{9} = 20°$$

Thus $a = 3(20°) = 60°$.
The four angles are $60°$, $60°+20° = 80°$, $60°+40° = 100°$, and $60°+60° = 120°$.

Four angles of the quadrilateral: **60°**, **80°**, **100°**, **120°**

Q9. Find three numbers in AP whose sum is 21 and the product of the first and third numbers exceeds the second number by 6. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Choose the three numbers symmetrically in AP:

Formula / Theorem Used: Three numbers in AP: (a - d), a, (a + d)

Step 1: Finding the Middle Number a [1 Mark]
Sum $= 21 \implies (a-d)+a+(a+d) = 21 \implies 3a = 21 \implies a = \frac{21}{3} = 7$.
Thus the numbers are $(7-d),\ 7,\ (7+d)$.

Step 2: Applying the Product Condition [1.5 Marks]
Given (First $\times$ Third) = Second $+6$:

$$(7-d)(7+d) = 7+6 \implies 49-d^2 = 13 \implies d^2 = 49-13 = 36 \implies d = \pm\sqrt{36} = \pm 6$$

Step 3: Listing the Three Numbers [0.5 Mark]
When $d = 6$: $7-6 = 1,\ 7,\ 7+6 = 13$. When $d = -6$: $13, 7, 1$.

Required numbers: **1**, **7**, **13** (or **13**, **7**, **1**)

Q10. CASE STUDY: Automobile Bank Loan Repayment Amortization

Rishi purchased a brand-new electric SUV and financed it with an auto-loan from a nationalized bank. To encourage disciplined early repayment, the bank offers a stepped repayment scheme. Rishi agrees to pay his total loan through monthly installments. He starts with a first installment of Rs. 10,000 and commits to increasing each subsequent monthly installment by Rs. 1,000 over the preceding month until the entire loan is completely discharged. [4]

(i) Write down the first three terms of the AP representing the monthly installments paid by Rishi. [1]

(ii) What amount does Rishi pay to the bank in his 10th monthly installment? [1]

(iii) What total cumulative amount has Rishi paid to the bank in the first 12 installments (1 full year)? [2]

— **OR** —

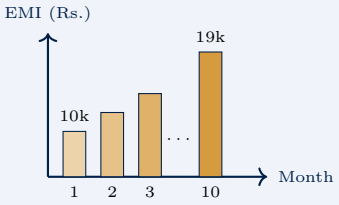
If the total loan principal amount to be cleared was Rs. 1,18,000, how many monthly installments will he need to clear the debt? [2]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Extracted Scenario Data: Monthly installments form an AP with first installment a = Rs. 10,000 and uniform monthly increase d = Rs. 1,000.

We know the general term and sum formulas of an AP:

Formula / Theorem
Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$



Sub-part (i): First Three Terms of the AP [1 Mark]

$a_1 = 10{,}000, \quad a_2 = 10{,}000 + 1{,}000 = 11{,}000, \quad a_3 = 11{,}000 + 1{,}000 = 12{,}000.$

Rs. **10,000**, Rs. **11,000**, Rs. **12,000**

Sub-part (ii): Amount of the 10th Monthly Installment [1 Mark]

$$a_{10} = 10{,}000 + (10-1)(1{,}000) = 10{,}000 + 9{,}000 = \boxed{\text{Rs. } \mathbf{19{,}000}}$$

Sub-part (iii): Total Amount Paid in First 12 Installments [2 Marks]

$$S_{12} = \frac{12}{2}[2(10{,}000) + 11(1{,}000)] = 6(20{,}000 + 11{,}000) = 6 \times 31{,}000 = \boxed{\text{Rs. } \mathbf{1{,}86{,}000}}$$

Sub-part (iii) [OR Option]: Number of Installments to Clear Rs. 1,18,000 [2 Marks]

Setting $S_n = 1{,}18{,}000$:

$$\frac{n}{2}[20{,}000 + (n-1)1{,}000] = 1{,}18{,}000 \implies \frac{n}{2}[1{,}000(19+n)] = 1{,}18{,}000 \implies n(n+19) = 236$$

$$n^2 + 19n - 236 = 0 \implies n = \frac{-19 + \sqrt{19^2 - 4(1)(-236)}}{2} = \frac{-19 + \sqrt{1305}}{2} \approx 8.562$$

*(Note: S_8 = Rs. 1,08,000 < Rs. 1,18,000 < Rs. 1,26,000 = S_9. Since $n \approx 8.56$ is not an integer—likely intended as S_8 = Rs. 1,08,000 in the paper—Rs. 1,08,000 is paid in 8 full installments and the remaining Rs. 10,000 clears the loan on the **9th installment**.)*

9 installments

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★